

Project Proposal Guidelines.

The proposal document must be in a PDF file, preferably generated by LaTeX, and it should be 1 page, plus an extra page for figures/tables and references. Your proposal should include the following sections (the order can be changed):

1. **Title:** Include a descriptive title and also the size of your group.
2. **Introduction or Problem statement:** What is the topic or goal of your project? Why do you think it is interesting or worth pursuing? Why do you think it is not trivial?
3. **Method:** Proposed approach or methods to be developed
 - (a) Explain the method (the high-level idea and optionally the technical part) in your own words, and background as much as needed (assuming the reader would have understood the topics covered in this class). A blind copy-and-paste is not allowed. Discuss the attributes of the method that warrant its use.
 - (b) Which machine learning methods are going to be investigated?
 - (c) Explain the method or approach you are developing. It is fine to not have a complete description.
4. **Related work:** Discuss related work. A comprehensive survey of related works is not necessary for the proposal. Elaborate on how cited references are different from or related to your project.
5. **Plan of Experiments:**
 - (a) **Implementation.** How will you implement your work? List the tools that will be used or developed.
 - (b) **Dataset to be used (real-world data or simulated).** You can discuss the nature of the dataset. How can it be obtained publicly, or how do you collect/generate the data? What are the statistics of the dataset? Why is it challenging, or why would it be useful/interesting to study?
 - (c) **Evaluation.** How will you evaluate the methods considered? What experimental settings, datasets, and evaluation metrics will you use?
6. **Plan of Project**
 - (a) Anticipated division of work over team.
 - (b) What challenges and difficulties would you expect? What are your contingency plans in case something goes wrong?
 - (c) Include milestones, to increase the likelihood that you are able to deliver at least some positive results.

Grading Guidelines.

- Writing quality: The document is well organized and clearly written. It is easy to follow and understand. Math notations and phrasing are accurate and technical. No obvious errors or typos.
- The problem under consideration is clearly stated and motivated.
- At least three key references are provided and their relevance to the project is explained.
- The methodologies and data analysis techniques are outlined at a high level.
- The evaluation process, including data sets, existing code repositories, and performance measures, are clearly described.
- The project plan is clearly stated, and describes how the work will be divided, milestones, and contingency plans if technical difficulties arise.
- The amount of work seems appropriate for the given group size